22ALT61 – TRANSFER LEARNING							
(Common to Artificial Intelligence and Machine Learning & Artificial Intelligence and Data Science branches)							
Programme & Branch	**B.Tech. Artificial Intelligence and Machine Learning & B.Tech. Artificial Intelligence and Data Science**	**Sem.**	**Category**	**L**	**T**	**P**	**Credit**
Prerequisites	**Machine Learning and Deep Learning**	**6**	**PC**	**3**	**0**	**0**	**3**
Preamble	This course enables to learn foundations of transfer learning and its applications						
Unit – I	**Transfer Learning Fundamentals:**						**9**
Introduction – Transfer learning strategies – Transfer learning and deep learning – Deep transfer learning types – Challenges of transfer learning – Need for transfer learning – Building CNN models – Leveraging transfer learning with pre-trained CNN models – Evaluating deep learning models.							
Unit – II	**Image Recognition and Classification:**						**9**
Deep learning based image classification – Benchmarking datasets – State-of-the-art deep image classification models – Image classification and transfer learning – Working with CIFAR-10 – Working with dog breed identification dataset.							
Unit – III	**Text Document Categorization:**						**9**
Traditional text categorization – Shortcomings of BoW model – Benchmark datasets – Word2vec model – Using gensim – GloVe model – Building a review sentiment classifier – Working with IMDB dataset with word embeddings – Creating document summaries using CNN model – Multiclass classification with CNN model – Visualizing document embeddings.							
Unit – IV	**Audio Event Identification and Classification:**						**9**
Understanding audio event classification – Exploratory analysis of audio events – Feature engineering and representation of audio events – Audio event classification with transfer learning – Building a deep learning audio event identifier.							
Unit – V	**Image Colorization:**						**9**
Problem statement – Color images – Building a coloring deep neural network – Preprocessing – Loss function – Encoder – Transfer learning – Feature extraction – Fusion layer – Decoder – Post-processing – Training and results – Challenges.							
							Total:45

TEXT BOOK:

1. Sarkar, D., Bali, R., & Ghosh, T., “Hands-On Transfer Learning with Python”, 1st Edition, Packt Publishing, 2018.

REFERENCES:

1. Qiang Yang, Yu Zhang, Wenyuan Dai, Sinno Jiallin Pan, “Transfer Learning”, 1st Edition, Cambridge University, 2020.
2. Jindong Wang, Yiqiang Chen, “Introduction to Transfer Learning”, 1st Edition, Springer, 2023.

COURSE OUTCOMES: On completion of the course, the students will be able to		**BT Mapped (Highest Level)**
CO1	utilize the basics of transfer learning to build CNN model	Applying (K3)
CO2	apply transfer learning techniques for image recognition and classification	Applying (K3)
CO3	experiment text document summarization using transfer learning	Applying (K3)
CO4	apply transfer learning for audio event identification and classification	Applying (K3)
CO5	perform image colorization using transfer learning	Applying (K3)

Mapping of COs with POs and PSOs

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2								3	2
CO2	3	2	2	2	2								3	2
CO3	3	2	2	2	2								3	2
CO4	3	2	2	2	2								3	2
CO5	3	2	2	2	2								3	2

1 – Slight, 2 – Moderate, 3 – Substantial, BT- Bloom's Taxonomy

ASSESSMENT PATTERN - THEORY

Test / Bloom's Category*	Remembering (K1) %	Understanding (K2) %	Applying (K3) %	Analyzing (K4) %	Evaluating (K5) %	Creating (K6) %	Total %
CAT1	-	50	50				100
CAT2	-	50	50				100
CAT3	-	50	50				100
ESE	-	50	50				100

* ±3% may be varied (CAT 1, 2, 3 – 50 marks & ESE – 100 marks)